Description of surface polaritons within a classical coupled harmonic oscillator model

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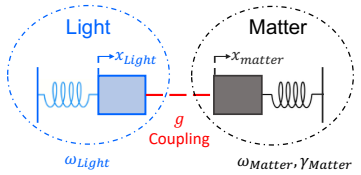
Abstract

Coupled harmonic oscillator models are widely used to describe strong and ultrastrong light–matter coupling. Surprisingly, this approach **has not been applied to analyse the surface phonon polariton (SPhP) and the surface plasmon polariton (SPP)**, which are among the most fundamental examples of ultrastrongly coupled light–matter modes. Here we apply the MoC model to revisit SPhPs and SPPs. We address **key questions** that are rarely discussed in the literature, including the **identification of the bare** (uncoupled) **light and matter modes** that hybridizes to form surface polaritons, the **complementary polariton branch**, and the **coupling strength**. We show that the surface polaritons are exactly reproduced when the uncoupled oscillators are chosen as the surface phonon (or surface plasmon) mode and the dielectric Brewster mode.

Coupled harmonic oscillator model



CHOM describe the interaction between oscillating system. Two bare mode couples with strength g to form two hybrid modes. When light and matter coupled the hybrid modes are called polaritons.

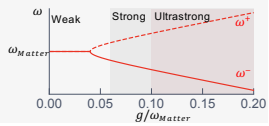
Momentum coupling (MoC) model

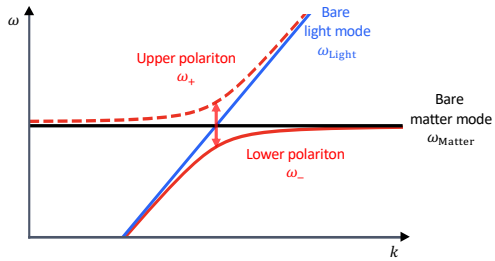
$$\ddot{x}_{\text{Light}} + \omega_{\text{Light}}^2 x_{\text{Light}} + 2g\,\dot{x}_{\text{Matter}} = 0$$

$$\ddot{x}_{\text{Matter}} + \gamma_{\text{Matter}}\dot{x}_{\text{Matter}} + \omega_{\text{Matter}}^2 x_{\text{Matter}} - 2g\,\dot{x}_{\text{Light}} = 0$$

Strong coupling
$g > |\gamma_{\text{Matter}}|/4$

Ultrastrong coupling
$g > 0.1\,\omega_{\text{Matter}}$

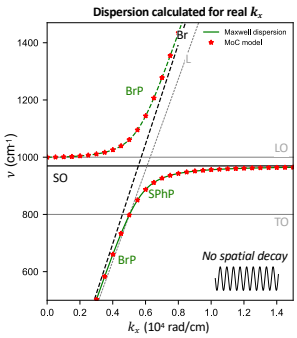




SURFACE PHONON POLARITON

We compared the dispersion obtained from Maxwell's equations to the MoC hybrid mode solution.

MoC model	$\omega_\pm^2 = \frac{1}{2}\left\{\left[\omega_{\text{Light}}^2 + \omega_{\text{Matter}}^2 + 4g^2\right] \pm \sqrt{\left[\omega_{\text{Light}}^2 + \omega_{\text{Matter}}^2 + 4g^2\right]^2 - 4\omega_{\text{Light}}^2\omega_{\text{Matter}}^2}\right\}$
Maxwell's dispersion	$\omega_\pm^2 = \frac{1}{2}\left\{\left[\omega_{\text{Br}}^2 + \omega_{\text{SO}}^2 + (\omega_{\text{LO}}^2 - \omega_{\text{SO}}^2)\right] \pm \sqrt{\left[\omega_{\text{Br}}^2 + \omega_{\text{SO}}^2 + (\omega_{\text{LO}}^2 - \omega_{\text{SO}}^2)\right]^2 - 4\omega_{\text{Br}}^2\omega_{\text{SO}}^2}\right\}$



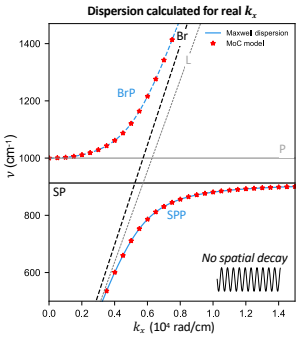
MoC model	Maxwell's dipersion
Hybrid Modes	
ω_+	ω_{BrP}
ω_-	$\omega_{SPhP}/\omega_{\text{BrP}}$
Bare Modes	
ω_{Light}	$\omega_{\text{Br}} = ck\sqrt{\frac{\varepsilon_1 + \varepsilon_\infty}{\varepsilon_1\varepsilon_\infty}}$
ω_{Matter}	$\omega_{\text{SO}} = \sqrt{\frac{\varepsilon_\infty\omega_{\text{LO}}^2 + \varepsilon_1\omega_{\text{TO}}^2}{\varepsilon_1 + \varepsilon_\infty}}$
Coupling strength	
g	$\sqrt{\frac{\omega_{\text{LO}}^2 - \omega_{\text{SO}}^2}{4}}$

Surface phonon polariton can be described by the ultrastrong coupling between the surface optical phonon resonance, ω_{SO}, and the dielectric Brewster mode, ω_{Br}. The lower polariton is the Surface phonon polariton (SPhP) while the upper polariton is the Brewster polariton (BrP).

SURFACE PLASMON POLARITON

We compared the dispersion obtained from Maxwell's equations to the MoC hybrid mode solution.

MoC model	$\omega_\pm^2 = \frac{1}{2}\left\{\left[\omega_{\text{Light}}^2 + \omega_{\text{Matter}}^2 + 4g^2\right] \pm \sqrt{\left[\omega_{\text{Light}}^2 + \omega_{\text{Matter}}^2 + 4g^2\right]^2 - 4\omega_{\text{Light}}^2\omega_{\text{Matter}}^2}\right\}$
Maxwell's dispersion	$\omega^2(k_x) = \frac{1}{2}\left\{\left[\omega_{\text{Br}}^2 + \omega_{\text{SP}}^2 + (\omega_{\text{P}}^2 - \omega_{\text{SP}}^2)\right] \pm \sqrt{\left[\omega_{\text{Br}}^2 + \omega_{\text{SP}}^2 + (\omega_{\text{P}}^2 - \omega_{\text{SP}}^2)\right]^2 - 4\omega_{\text{Br}}^2\omega_{\text{SP}}^2}\right\}$

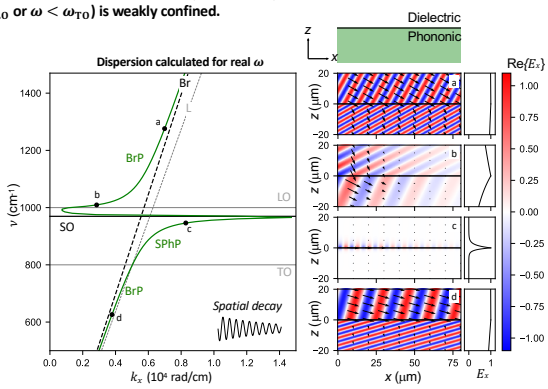


MoC model	Maxwell's dipersion
Hybrid Modes	
ω_+	ω_{BrP}
ω_-	ω_{SPP}
Bare Modes	
ω_{Light}	$\omega_{\text{Br}} = ck\sqrt{\frac{\varepsilon_1 + \varepsilon_\infty}{\varepsilon_1\varepsilon_\infty}}$
ω_{Matter}	$\omega_{\text{SP}} = \sqrt{\frac{\varepsilon_\infty\omega_{\text{P}}^2}{\varepsilon_1 + \varepsilon_\infty}}$
Coupling strength	
g	$\sqrt{\frac{\omega_{\text{P}}^2 - \omega_{\text{SP}}^2}{4}}$

Surface plasmon polariton can be described by the ultrastrong coupling between the surface plasmon resonance, ω_{SP}, and the dielectric Brewster mode, ω_{Br}. The lower polariton is the surface plasmon polariton (SPP) while the upper polariton is the Brewster polariton (BrP).

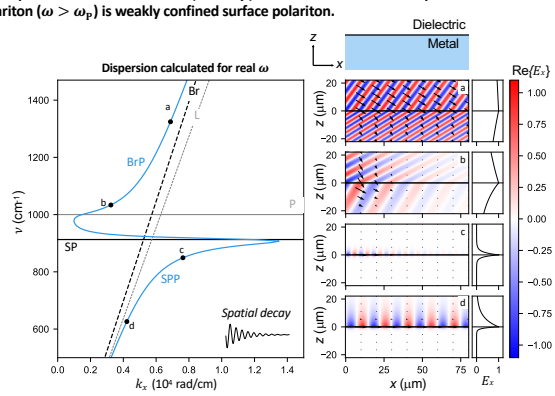
Mode profile

Electric field $Re[E_x]$ profile and time average Poynting vector at different points along the surface polariton dispersion. The field always propagate along the surface and decays exponentially in the out-of-plane direction. The SPhP ($\omega_{\text{TO}} < \omega < \omega_{\text{LO}}$) is ultraconfined while the Brewster polariton ($\omega > \omega_{\text{LO}}$ or $\omega < \omega_{\text{TO}}$) is weakly confined.



Mode profile

Electric field $Re[E_x]$ profile and time average Poynting vector at different points along the surface polariton dispersion. The field always propagate along the surface and decays exponentially in the out-of-plane direction. The SPP ($\omega < \omega_P$) is ultraconfined surface polariton while the Brewster polariton ($\omega > \omega_P$) is weakly confined surface polariton.



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